

Properties of the Cosmic Web ab initio

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July 1st 2026

The Random Universe: Imperial College London,

Andrew Jaffe @ 60

June 29 – July 2, 2026

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Cosmology from MAXIMA-1, BOOMERANG & COBE/DMR CMB Observations.

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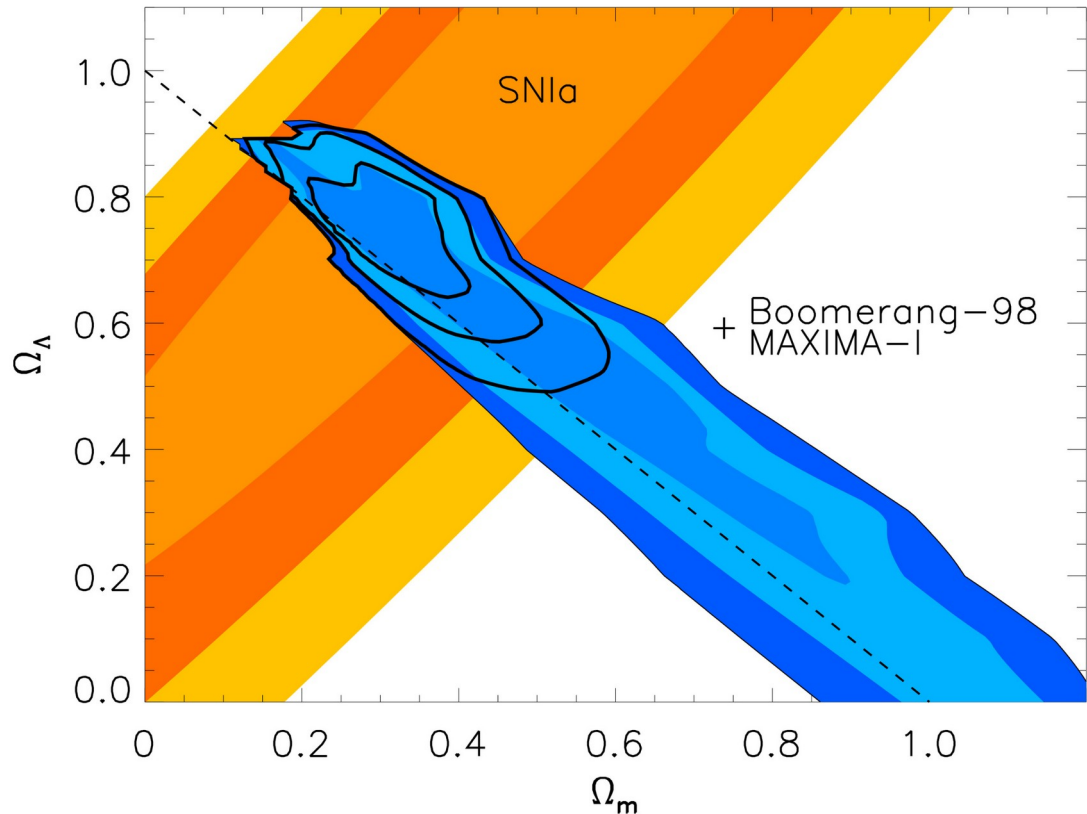
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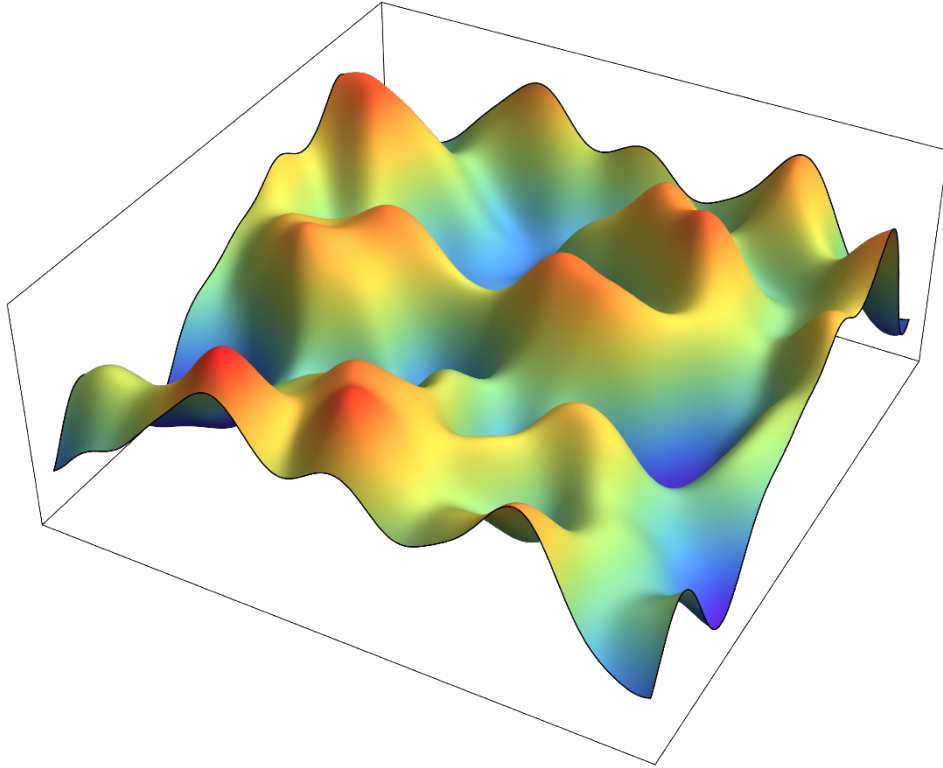
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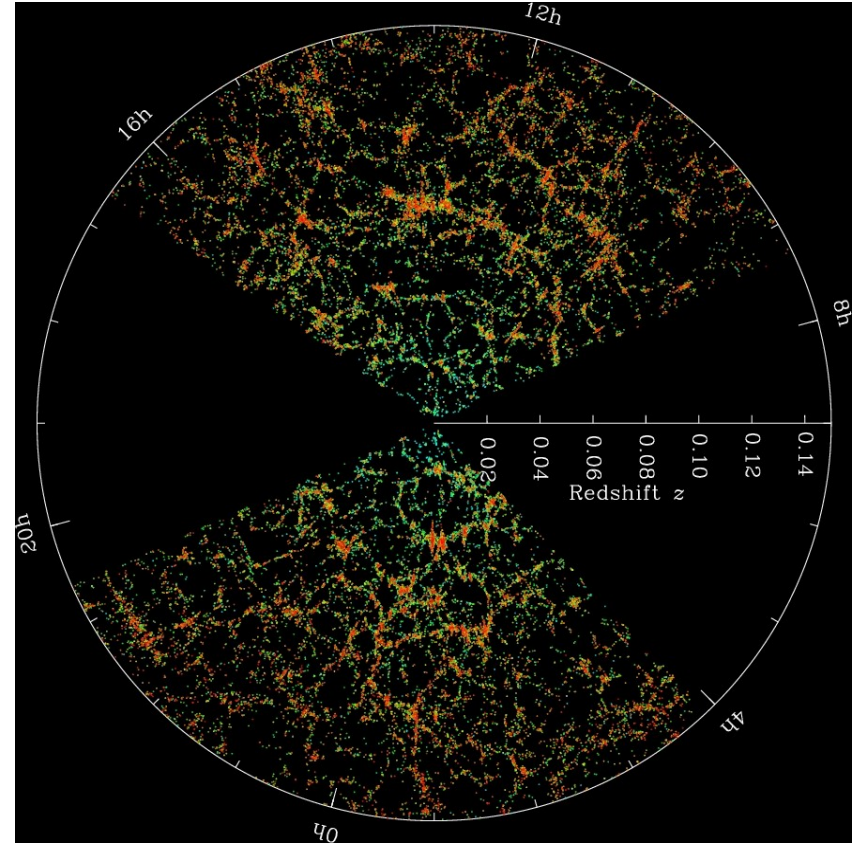
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“ab initio” – understand structure from initial field



2D for illustration, initial density
field fully describes initial conditions

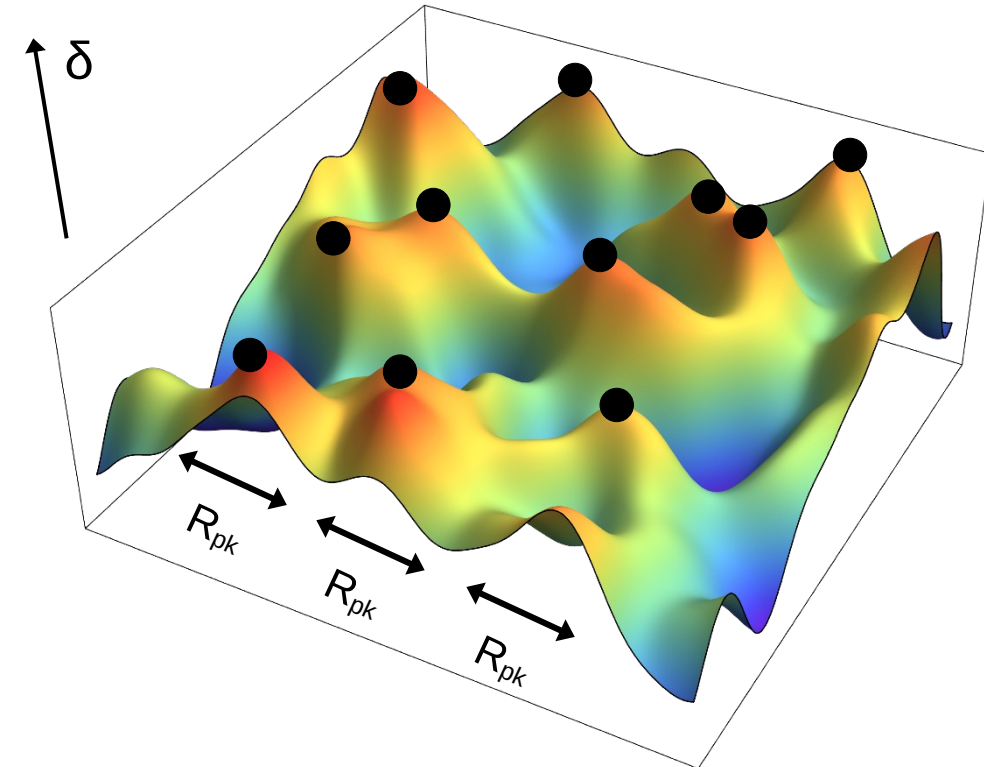


Sloan DS Survey

Peak (maxima) – Halo correspondence

(Bardeen, Bond, Kaiser, Szalay 1986)

Fixed smoothing R_{pk} picture $\neq$ constant time slice



Smoothed initial density

Halo mass $M \propto$ peak scale R_{pk}^3

Height δ_{pk} of the peak gives time when this mass collapses (i.e when this halo exists)

$$\delta_{pk} D(a) = 1.681$$

$$D(a) = 1.681 / \delta_{pk}$$

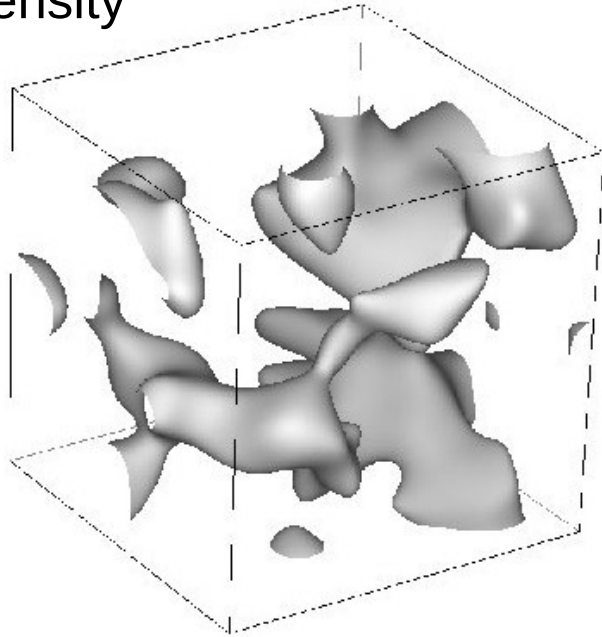
(spherical collapse model)

$D(a)$ is growing mode of perturb
as function of scale factor a

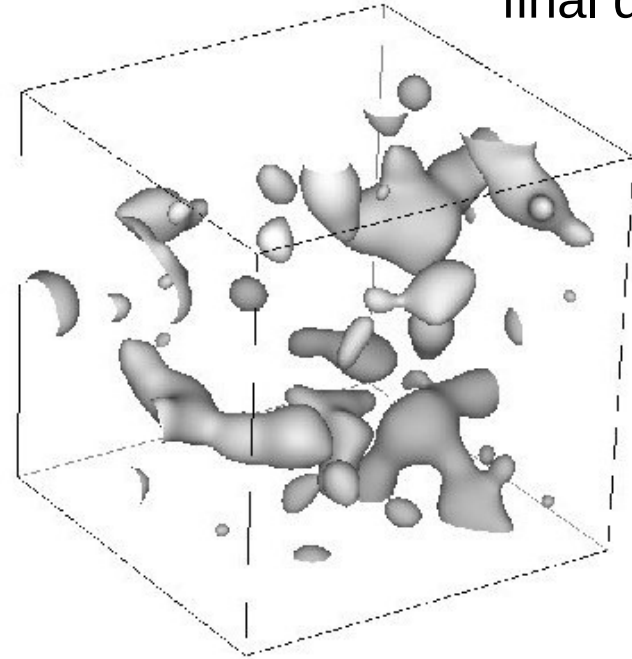
Filamentary Cosmic Web

(Bond, Kofman, Pogosyan, 1996)

initial linear density



final density



Not only peak of initial density field can be found as halos in the final structures
But at lower overdensity thresholds, filamentary ridges between peaks in initial fields are also mapped into filamentary bridges between halos of final LSS at mildly non-linear scales.

Isle of Skye: 2D peaks and ridges



Image credit: Zoltan Gabor, <http://zoltangabor.com>

Connectivity of the fully connected skeleton

(Codis,Pogosyan,Pichon,2018)

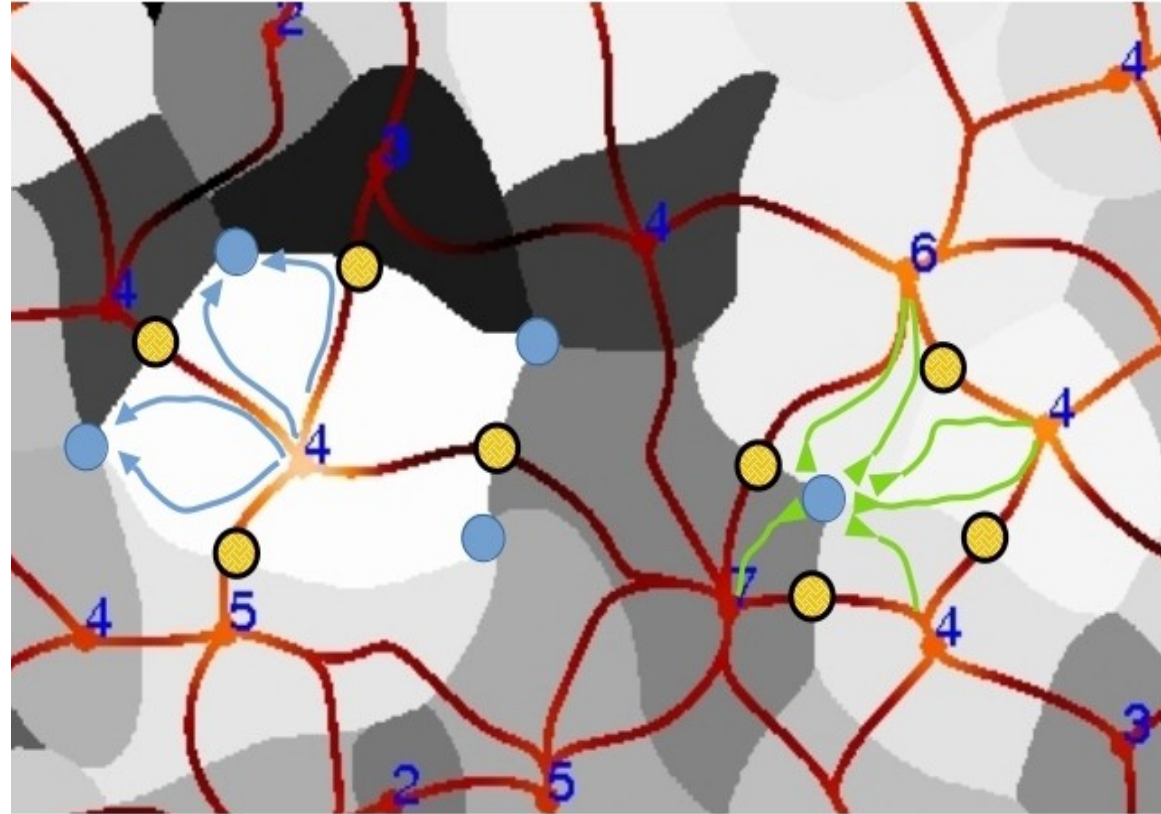
Connectivity κ is number of peaks the given peak is connected to

κ as the function of peak properties, and/or restrictions put on saddles is the main object of interest

Without any restrictions, an immediate estimate is

$$\langle \kappa \rangle \approx 2 \frac{\langle n_{saddles} \rangle}{\langle n_{peak} \rangle} \approx 6.1$$

but not every segment of the skeleton is a dense filament, but all will be important for merger history.



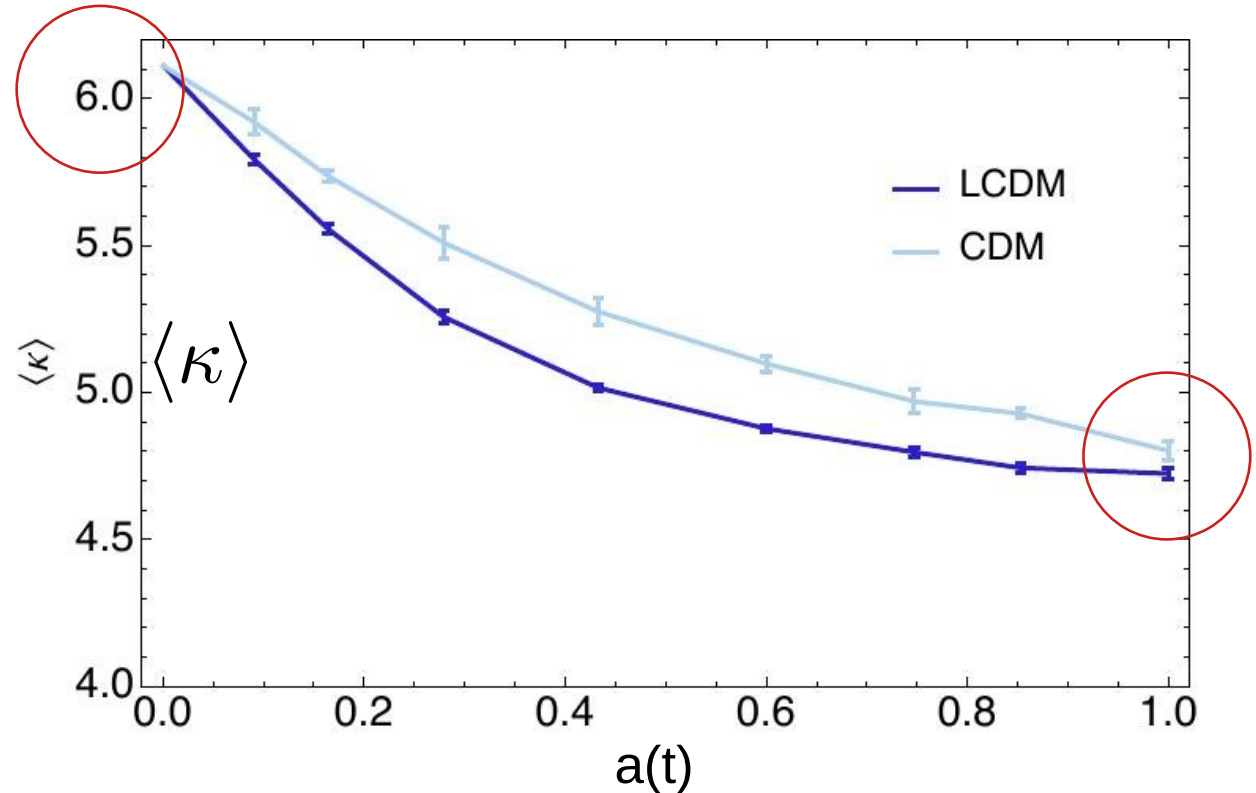
Let us run simulations and **measure connectivity of density peaks**

Fixed comoving Gaussian
smoothing with $R=10\text{Mpc}$

Field is mildly nonlinear
and non-Gaussian at
present time.

“timing” of the curve track
the evolution of the
growing mode factor and
cosmology dependent

(Codis et al, 2018)

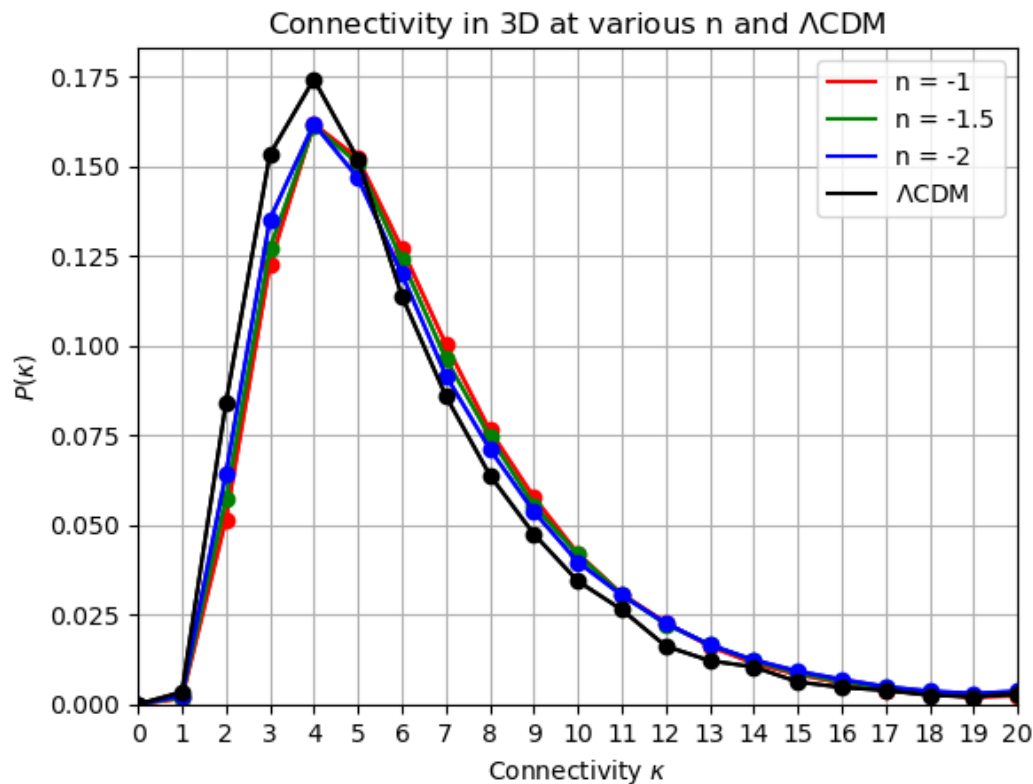


Linear, Gaussian

$\sigma \approx 0.2$
a bit non-Gaussian, also predicted,
(Gay, Pichon, Pogosyan, 2012)

Probability distribution of connectivity from measurement on Gaussian fields with different spectra

Aahan Pahwa, 2025

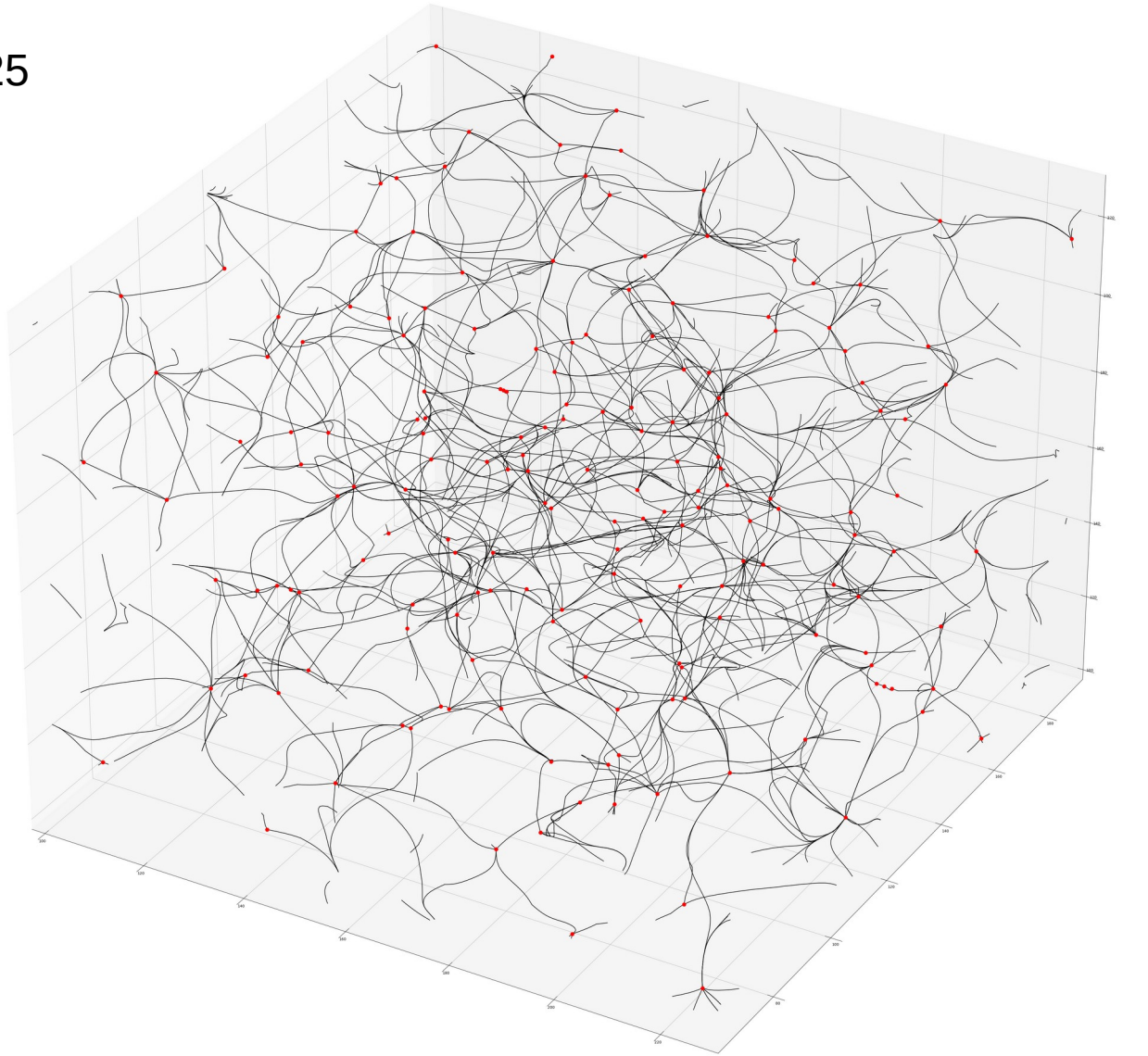


Notable results: $\langle \kappa \rangle \approx 6$ but most probable value is $\kappa_{mod} \approx 4$

Aahan Pahwa, DP, 2025

Connectivity of LCDM

All skeleton filaments

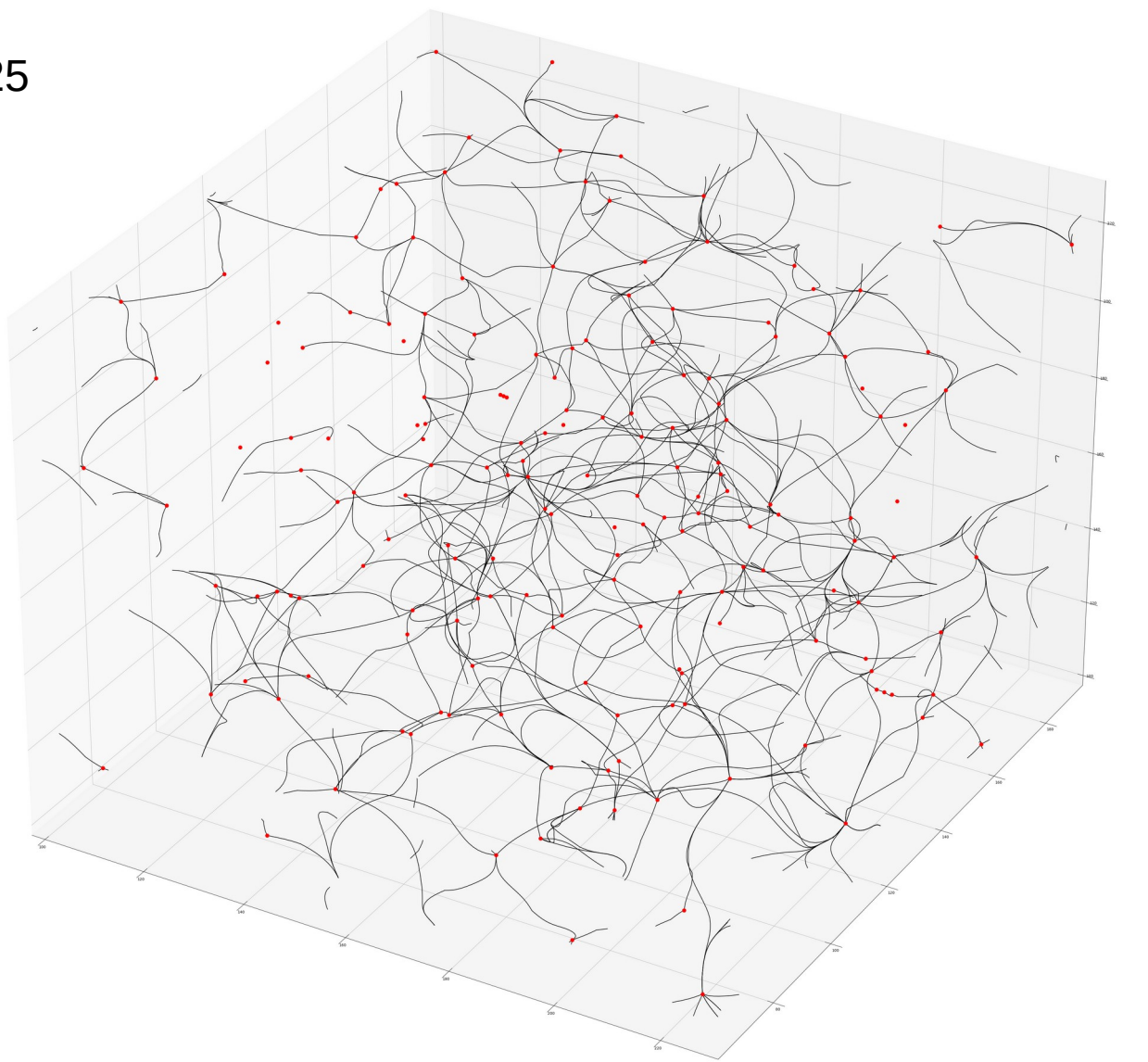


Aahan Pahwa, DP, 2025

Connectivity of LCDM

Overdense filaments

$$\delta_{fil} = \delta_{saddle} > 0$$

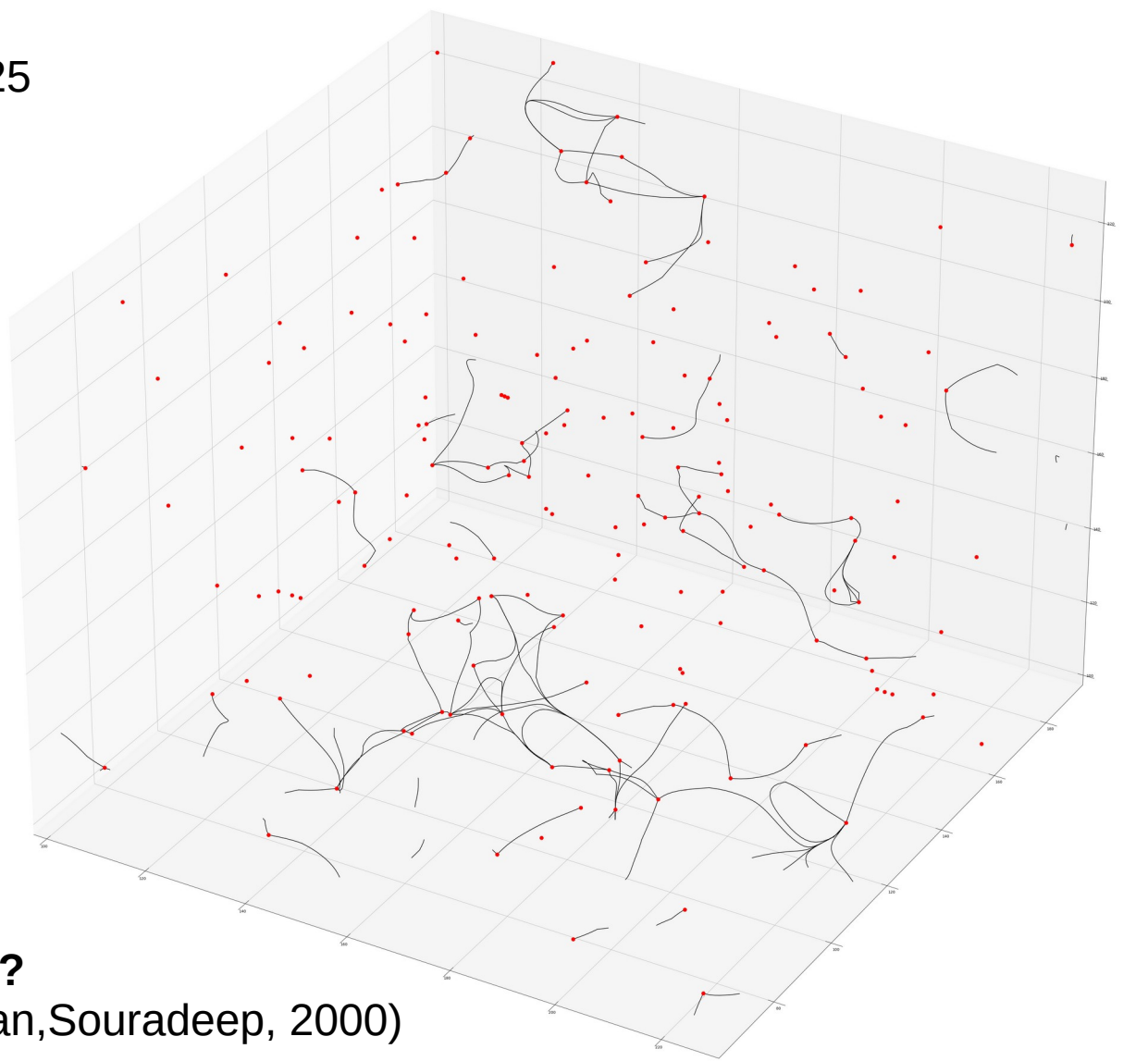


Aahan Pahwa, DP, 2025

Connectivity of LCDM

Overdense filaments

$$\delta_{fil} > 1.25\sigma$$



Percolation threshold ?
(recall Colombi, Pogosyan, Souradeep, 2000)

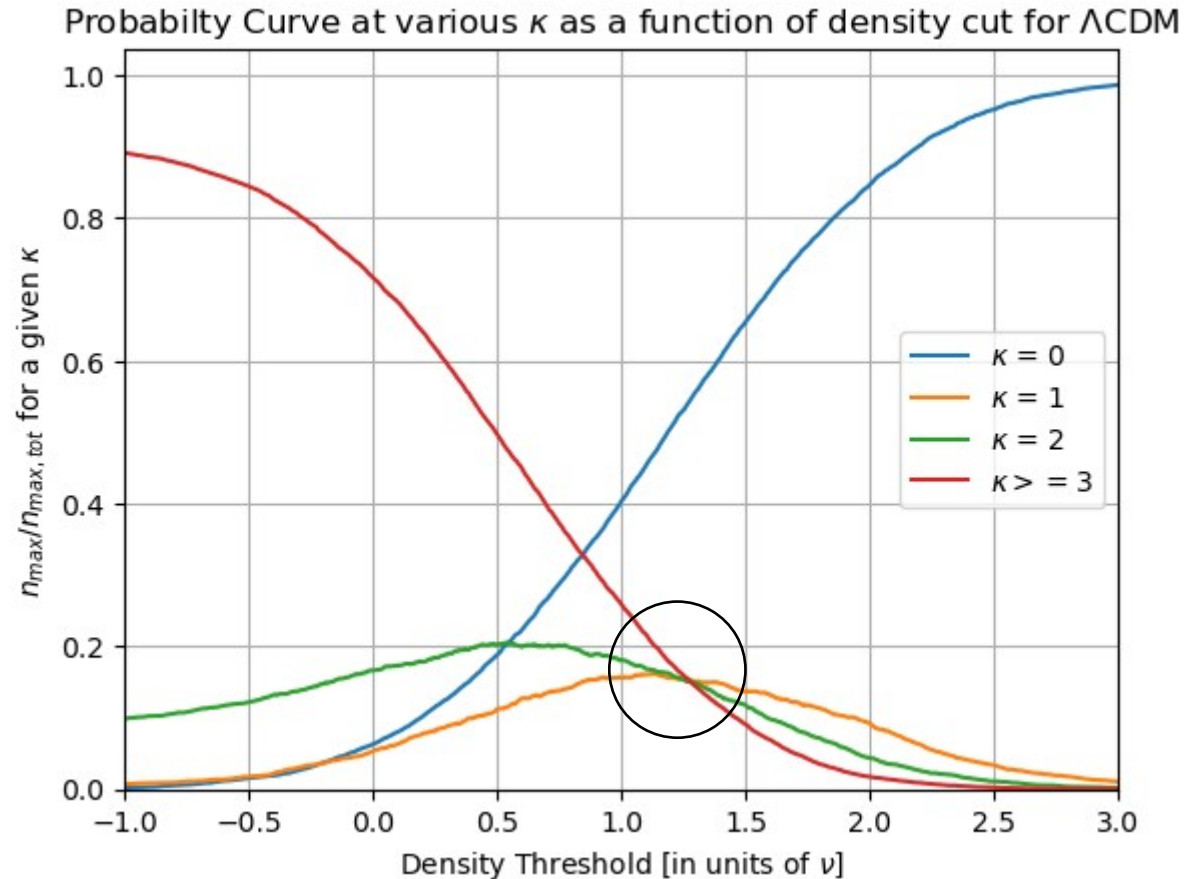
Fraction of peaks with particular connectivity as function of saddle point density in units of σ

Aahan Pahwa, DP, 2025

Connectivity of LCDM

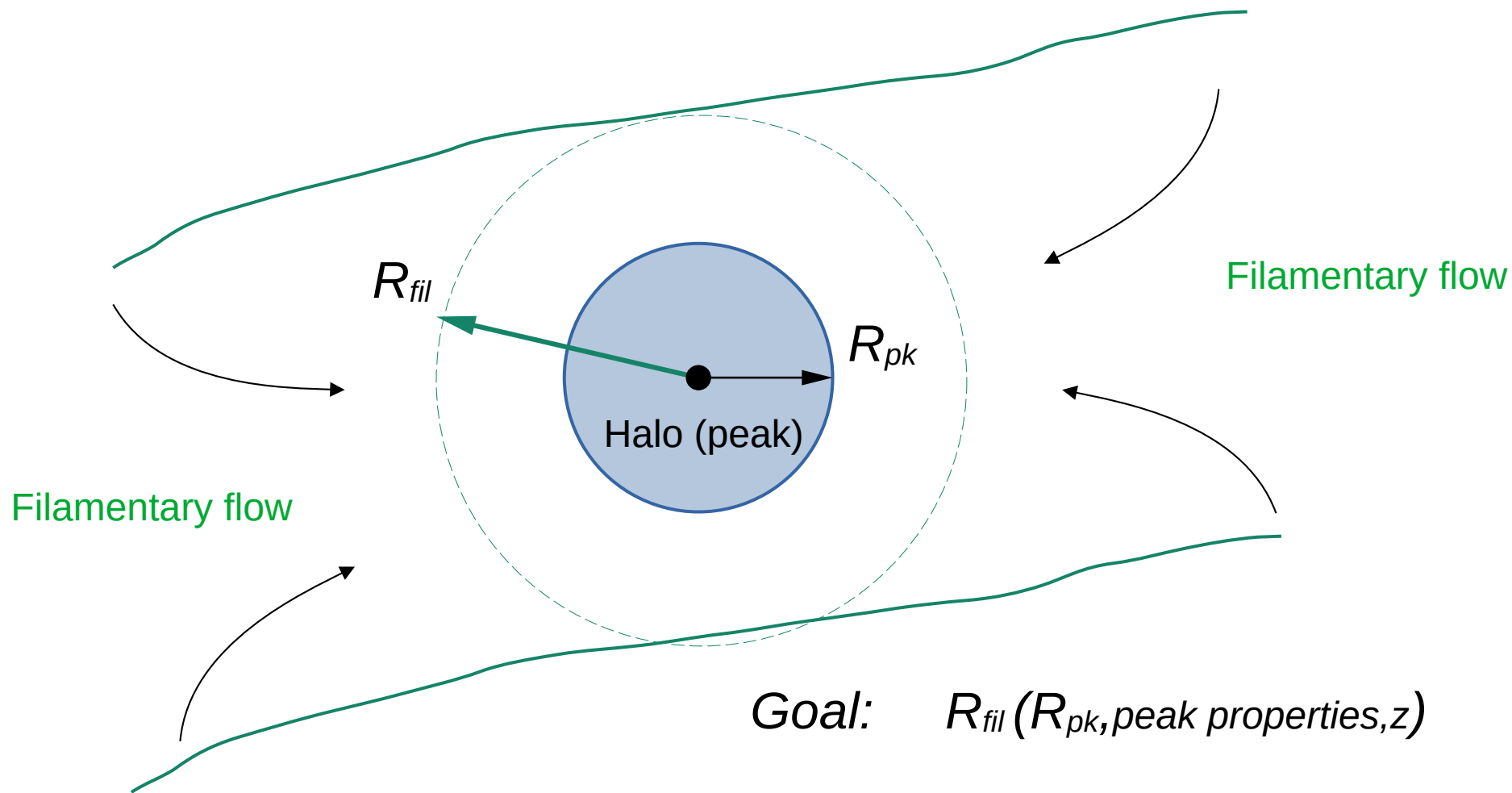
Fraction of halos that are:

Blue - unconnected
Orange - 1 connection
Green - 2 connections
Red - 3 or more



Scale of the filaments around collapsing halos

with Junsup Shim (ASIAA, Pusan U), Christophe Pichon (IAP), 2026



Simple(st) mathematical model of conditional distribution

$$P(\underbrace{\lambda_1, \lambda_2, \lambda_3}_{\text{Nature of surrounding flow}} \mid \underbrace{\delta}_{\text{Properties of the halo}})$$

Nature of surrounding flow

Properties of the halo

Eigenvalues of the strain tensor $\frac{\partial s_i}{\partial x_j}$
smoothed at R_{fil} scale

Density within spherical proto-halo of
scale R_{pk}

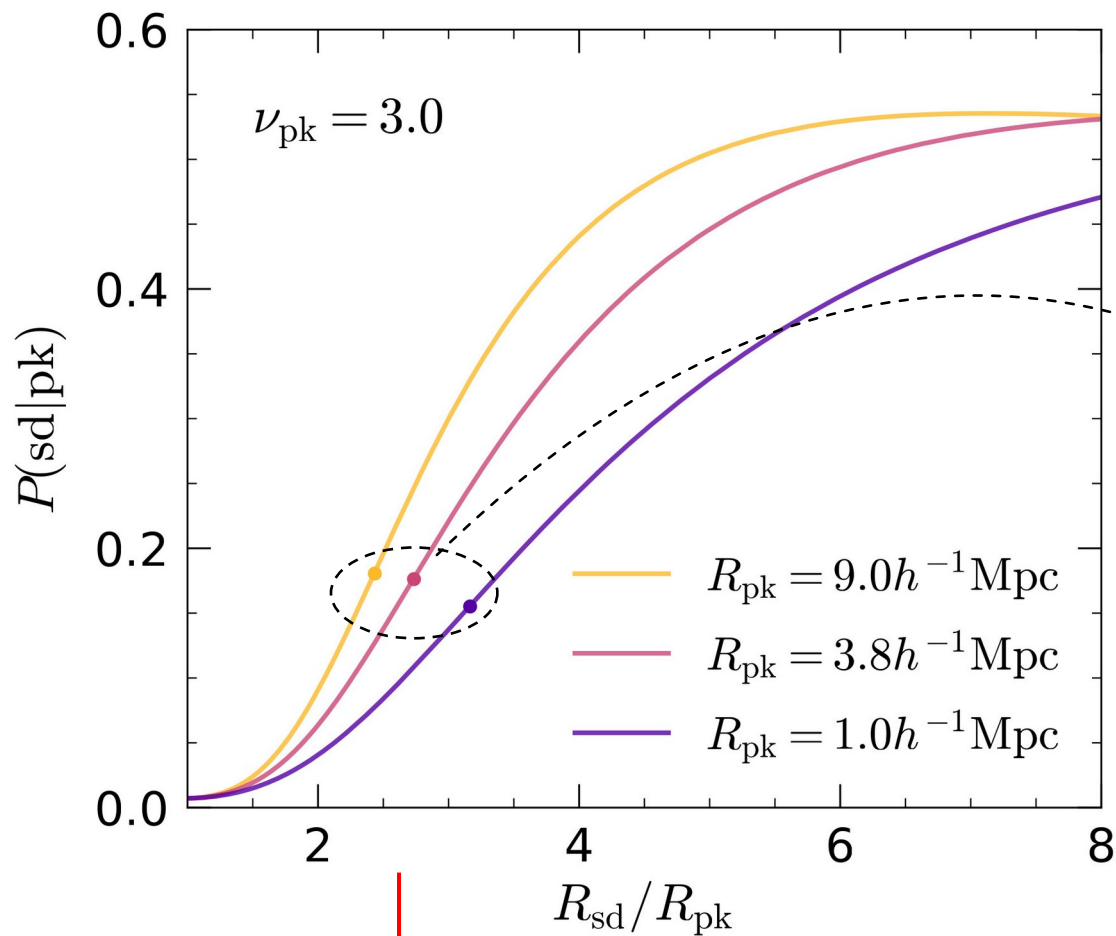
Require filamentary flow:

$$\lambda_3 < \lambda_2 < 0, \quad \lambda_1 \geq 0$$

Require high overdensity

$$\nu_{pk} = \delta / \sigma(R_{pk}, z) > 2$$

(ν_{pk} is “rarity”)



$\Lambda\text{CDM}, z=0$

$$\frac{R_{fil}}{R_{pk}} = f(R_{pk}, \nu_{pk})$$



Spherical flow around the peak



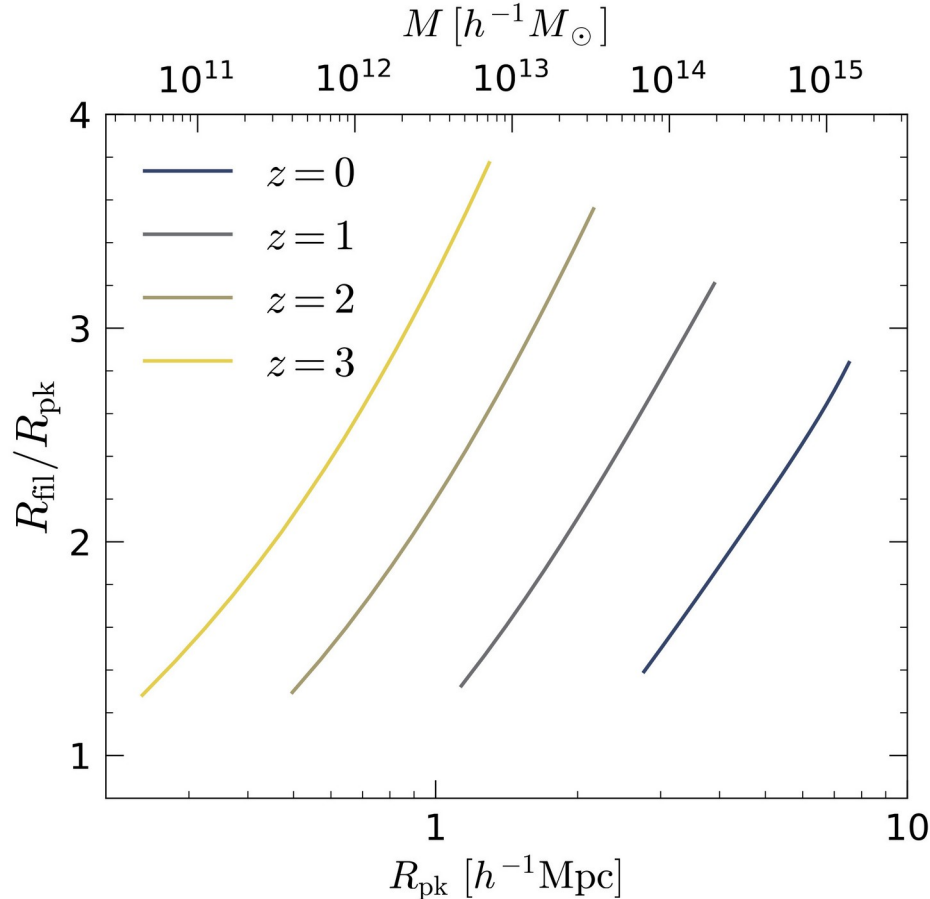
Global mean probability of filamentary region is 0.42

Filamentary environment becomes important for the halo around the inflection point

Filamentary environment of halos of different mass collapsing at a given redshift z

Resolving spherical collapse condition for ν_{pk}

$$\delta \equiv \nu_{pk} \sigma(R_{pk}) D(z) = 1.686 \quad \Rightarrow \quad \nu_{pk} = \frac{1.686}{\sigma(R_{pk}) D(z)}$$



$$(R_{\text{fil}}/R_{\text{pk}}) \approx (R_{\text{pk}}/1.6h^{-1}\text{Mpc})^{0.68} \quad z=0,$$

$$(R_{\text{fil}}/R_{\text{pk}}) \approx (R_{\text{pk}}/0.67h^{-1}\text{Mpc})^{0.68} \quad z=1,$$

$$(R_{\text{fil}}/R_{\text{pk}}) \approx (R_{\text{pk}}/0.32h^{-1}\text{Mpc})^{0.68} \quad z=2,$$

$$(R_{\text{fil}}/R_{\text{pk}}) \approx (R_{\text{pk}}/0.15h^{-1}\text{Mpc})^{0.62} \quad z=3,$$

$$R_{\text{fil}}(M) \approx 4.9 M_{14}^{0.56} h^{-1}\text{Mpc} \quad z=0,$$

$$R_{\text{fil}}(M) \approx 2.4 M_{13}^{0.56} h^{-1}\text{Mpc} \quad z=1,$$

$$R_{\text{fil}}(M) \approx 1.1 M_{12}^{0.56} h^{-1}\text{Mpc} \quad z=2,$$

$$R_{\text{fil}}(M) \approx 0.5 M_{11}^{0.54} h^{-1}\text{Mpc} \quad z=3.$$

The environment of the most prominent halos (high ν_{pk})

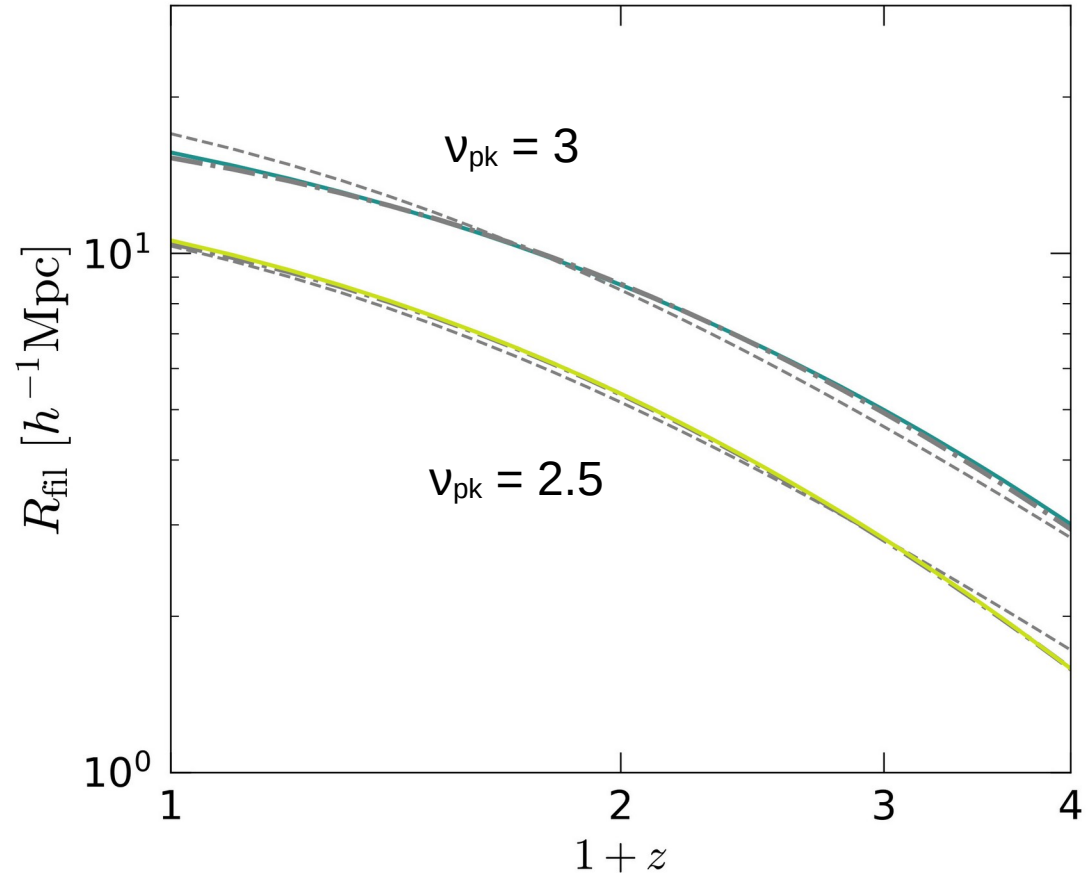
Fix $\nu_{pk} = \{2.5, 3, \dots\}$

Resolve collapse condition to relate R_{pk} to redshift z

$$\sigma(R_{pk}) = \frac{1.686}{\nu_{pk} D(z)}$$

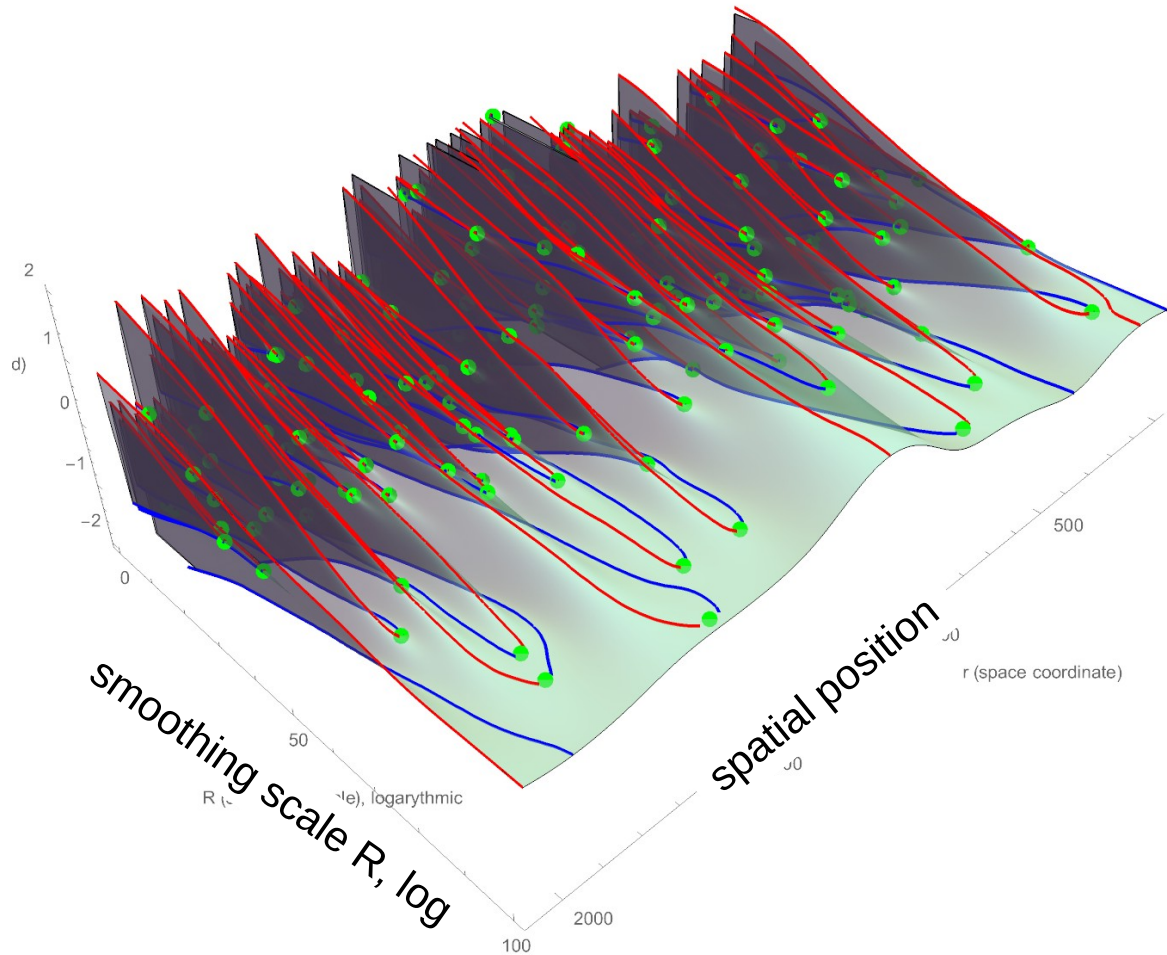
to obtain $R_{fil}(z)$

$$R_{fil} = \begin{cases} \frac{31 h^{-1} \text{Mpc}}{2 + (1+z)^2}, & \nu_{pk} = 2.5 \\ \frac{51 h^{-1} \text{Mpc}}{2 + (1+z)^2}, & \nu_{pk} = 3.0 \end{cases}$$



Density field transformation in extended position — smoothing scale ND+1 space (Hanami, 2001, Cadiou et al, 2020)

ND+1 space, density, N=1, Gaussian filter

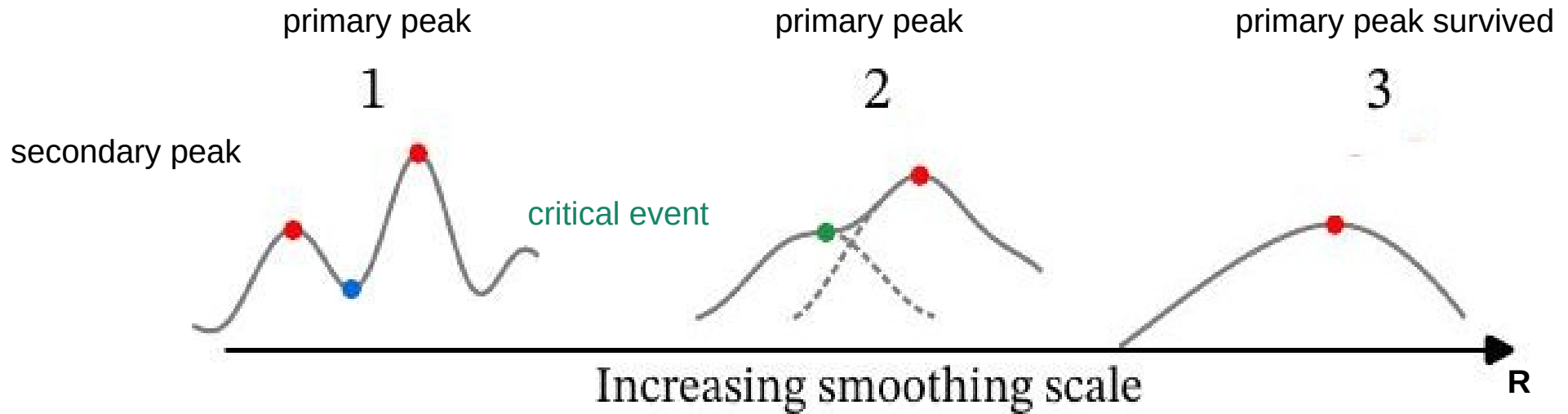


For Gaussian smoothing depolyed density δ falls monotonically with R along peak lines

As smoothing increases **peaks disappear, but some survive.**
Critical events are points in position-smoothing ND+1 space where peaks disappear.

The density and correlations of those points we studied in Cadiou et al 2020.

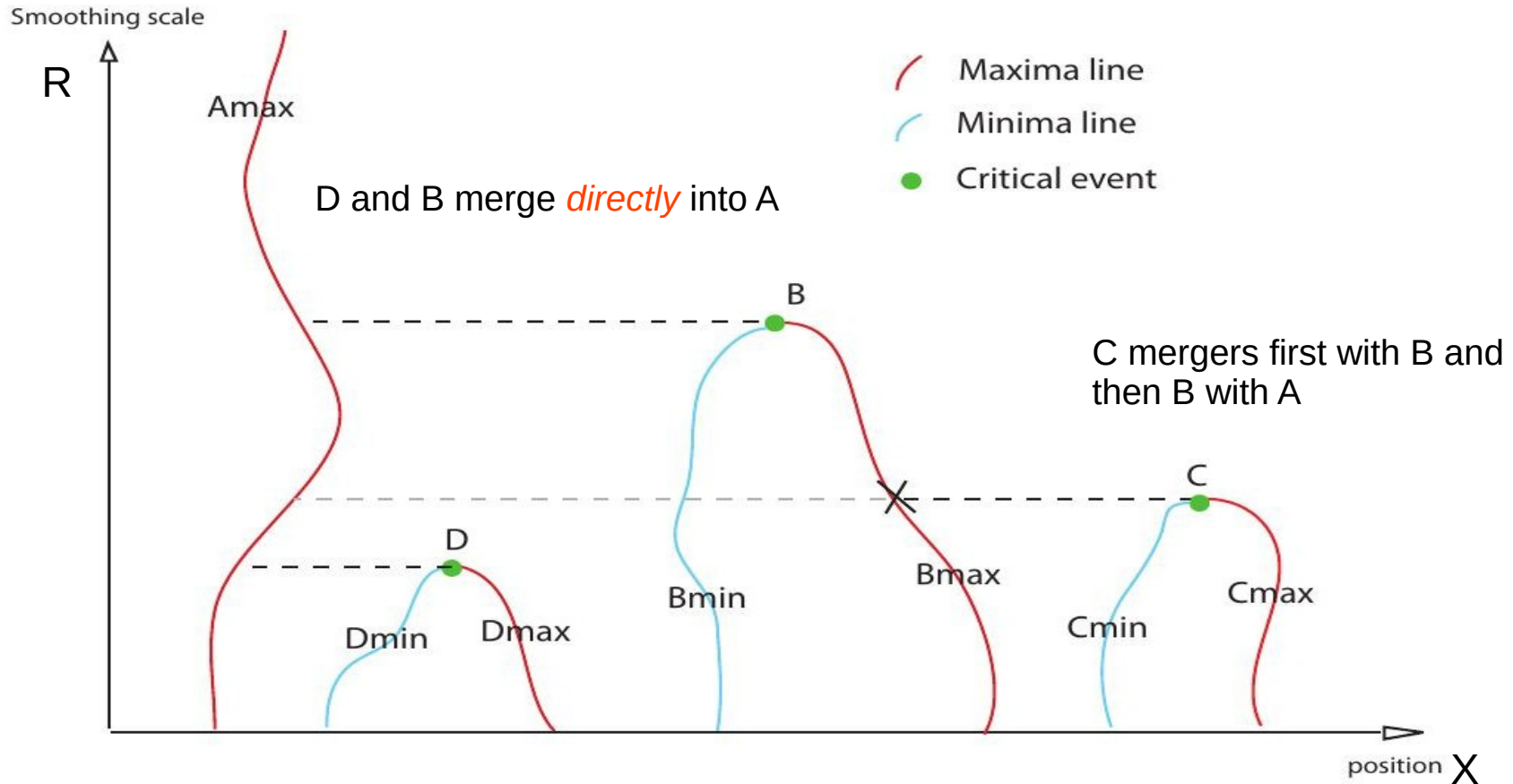
When smoothing is increased two peaks merge through **the critical event**



Critical event is the moment of coalescence of the secondary peak and the connecting filamentary saddle (in 1D – min)

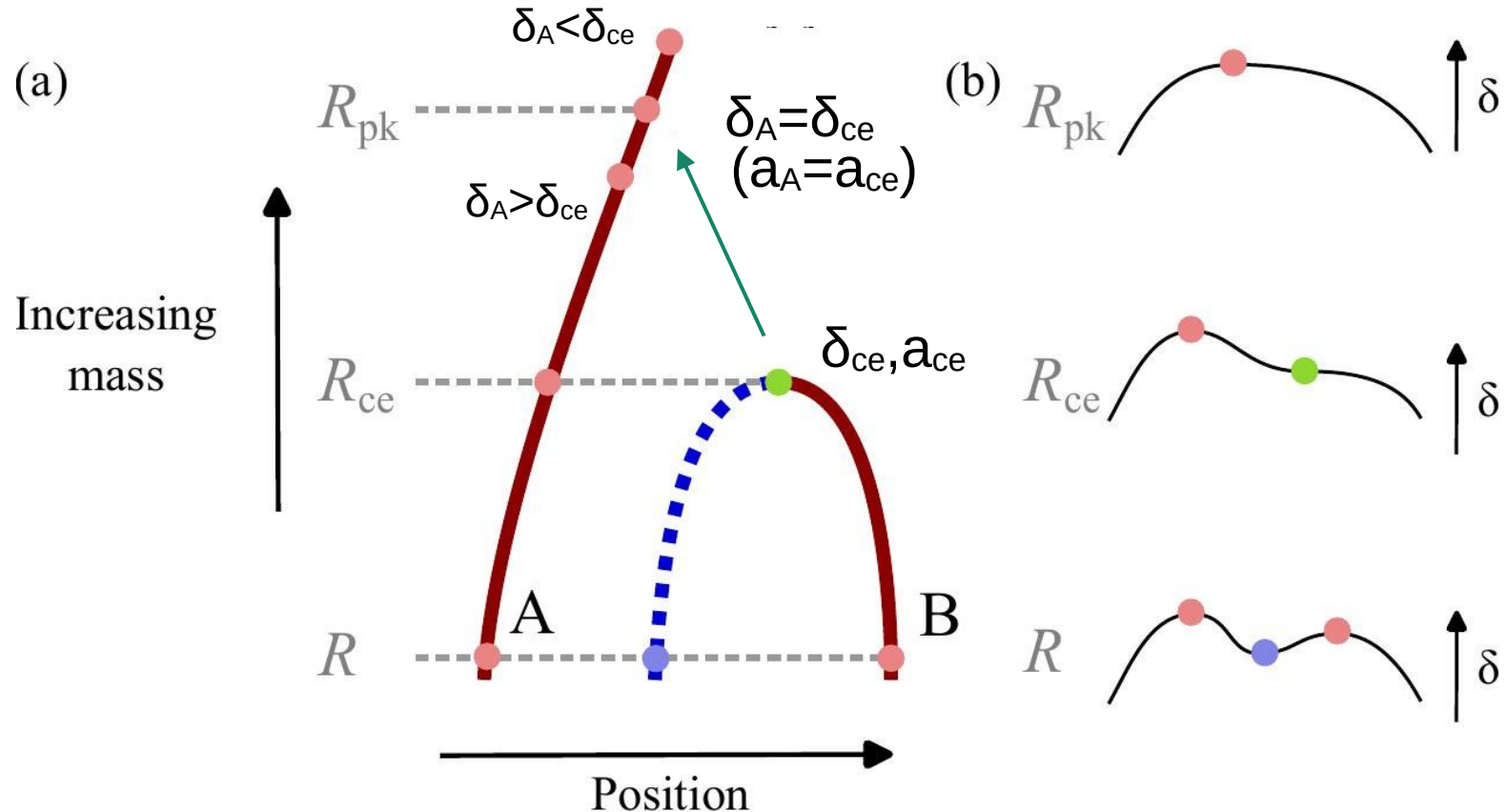
Gaussian smoothing – height of peaks decreases monotonically

Peaks and saddles track lines in extended position — smoothing scale ND+1 space



Critical events are points in position-smoothing scale ND+1 space where histories of one peak and one saddle end.

Peak-Halo association using spherical collapse model



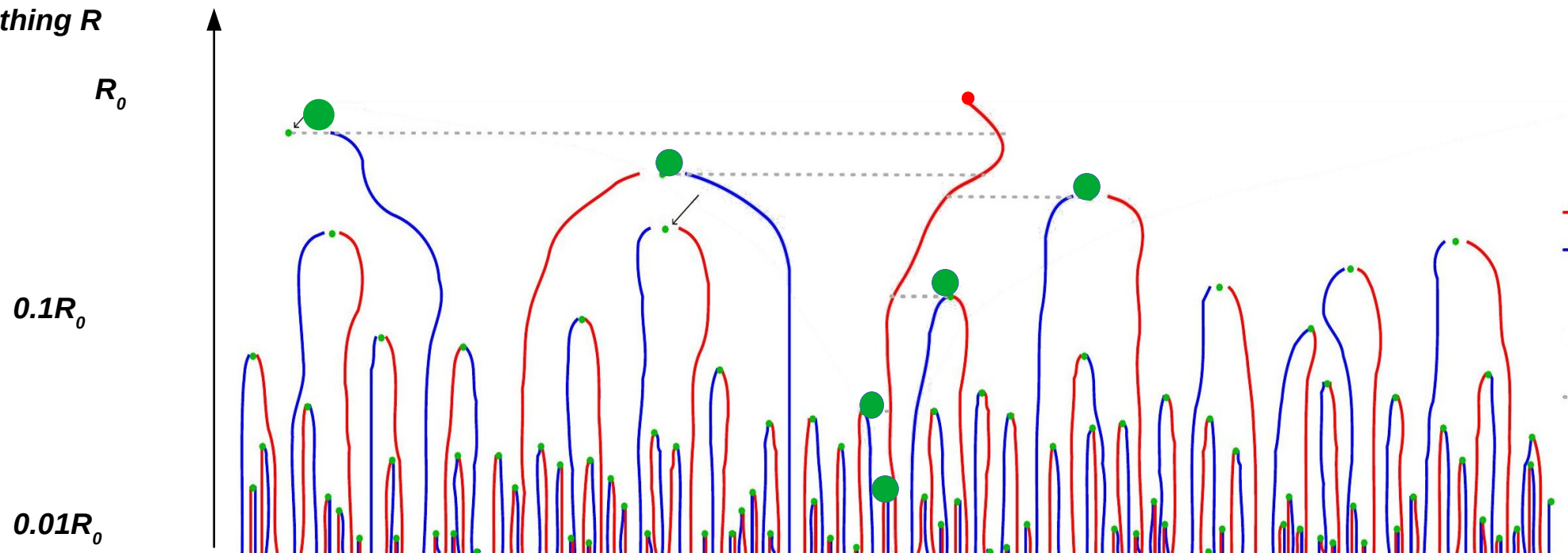
Central halo A of mass R_{pk} merges with satellite B of mass R_{ce} .

Implicit condition to find R_{pk} is $\delta_A(R_{pk}) = \delta_B(R_{ce})$ which is equivalent to requirement that these two halos are at the same moment $D(a_A) = D(a_B)$

Halo merger history through critical events

(Cadiou, Pharabot, Pichon, Pogosyan, 2024)

Smoothing R



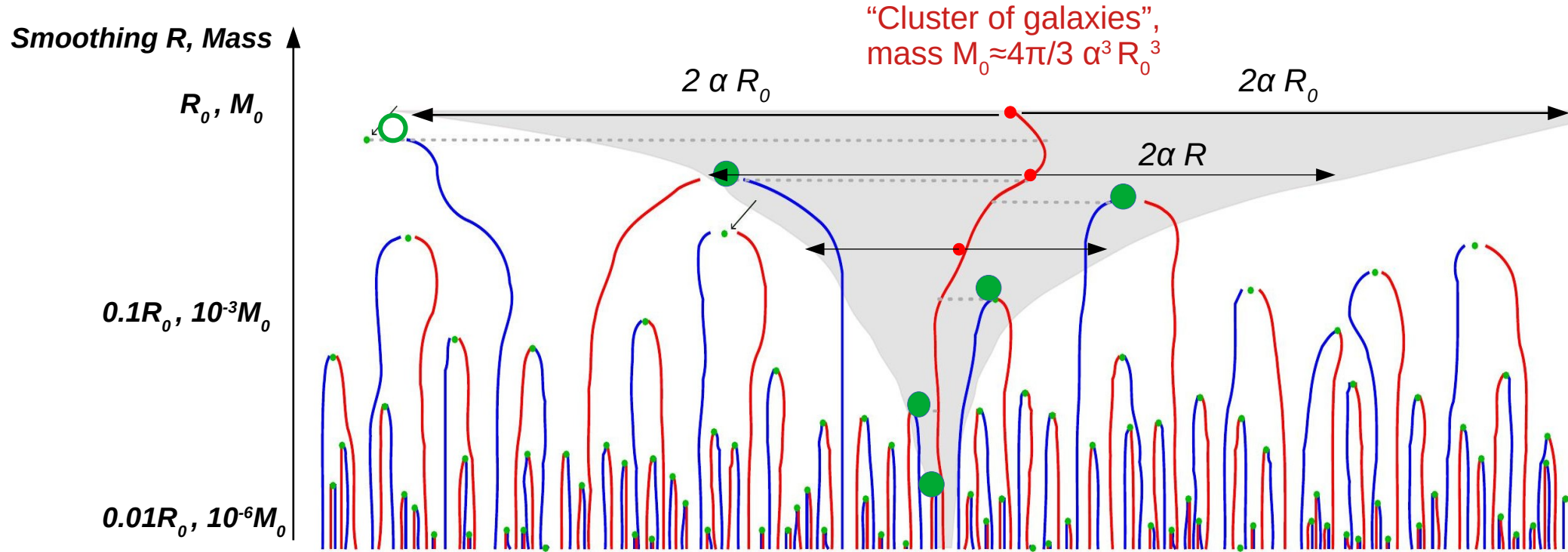
Points of note:



marks critical events that merger directly in the central peak

Halo past mass accumulation cone, merger capture cone

(Cadiou, Pharabot, Pichon, Pogosyan, 2024)



- Two regimes, a) monotonic accretion, b) merging crossing critical event
- **Points of note:** Just several mergers as mass goes from 10^{-6} to 1 (say $10^9 M_\odot$ to $10^{15} M_\odot$)
- $\alpha R = R_{\text{TH}}$ is the “effective top-hat size” for Gaussian filter R , $\alpha \approx 1.56$
- Critical events are near the boundary of the “merger capture cone” which is approx twice in radius to past mass cone. This argues for $\beta \approx 2\alpha \approx 3.1$

Merger number, in “zero” order

$$N_m = \int_{R_{in}}^{R_0} \underbrace{V_{capture}(R) dR}_{\text{R slice trough past cone}} \underbrace{n_{ce}(R|\text{conditions})}_{\substack{\text{4D density of critical events} \\ \text{Correlations with peaks, ...}}} .$$

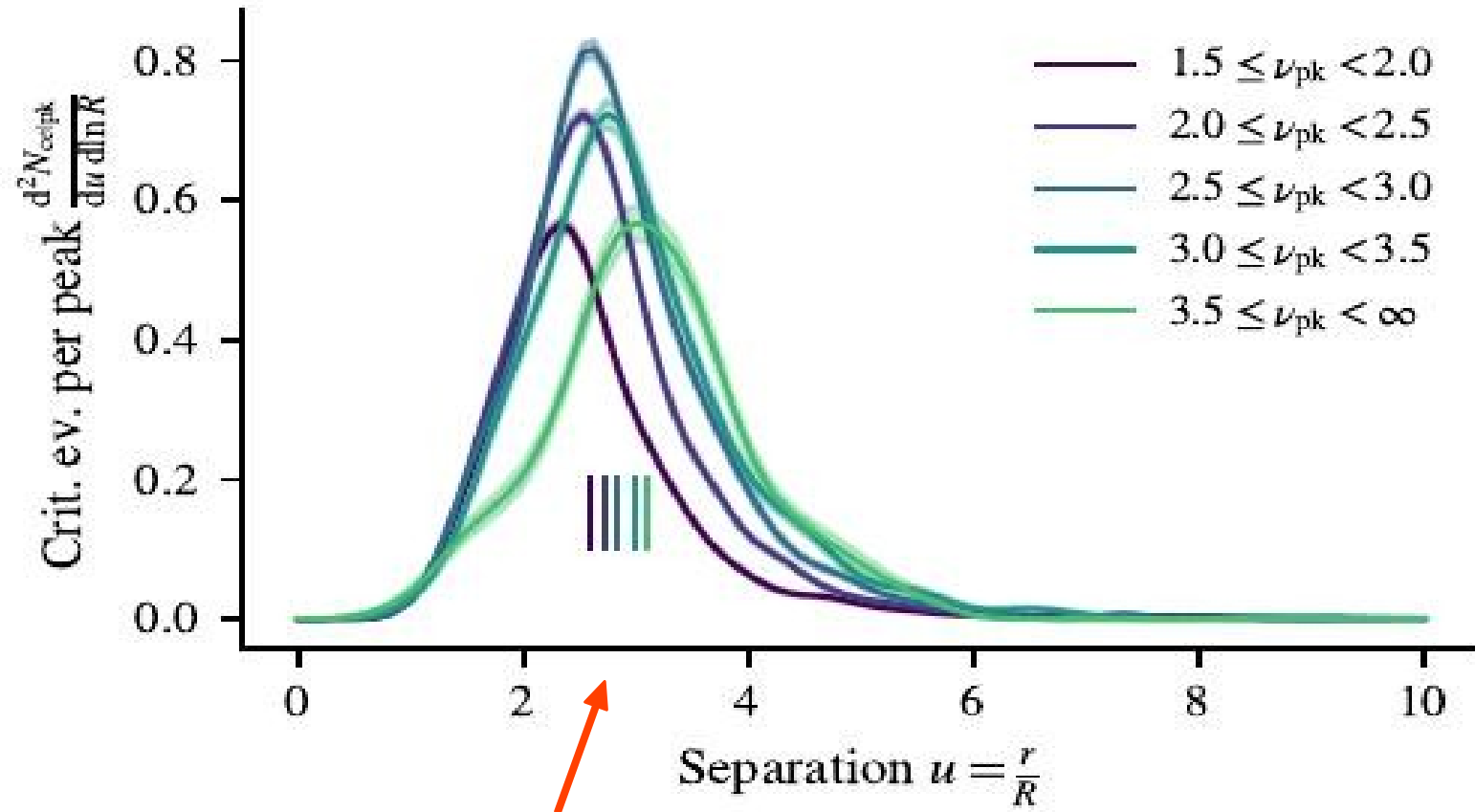
Ignore conditions, take mean density of critical events (Cadiou et al, 2020)

$$\bar{n}_{ce}(R) = \frac{3R}{R_*^5} \frac{1 - \tilde{\gamma}^2}{\tilde{\gamma}^2} \frac{29\sqrt{15} - 18\sqrt{10}}{1800\pi^2} \propto \frac{1}{R^4}$$

Capture volume is estimated as $V_{cone}(R) \approx 4\pi/3 (\beta R)^3$, take power-law spectrum with index n for spectral parameters $R_*, \tilde{\gamma}$, and relate $R \sim M^{1/3}$

$$N_m = \underbrace{0.9}_{\text{Order of 1}} \left(\frac{5+n}{2} \right)^{3/2} \left(\frac{\beta}{3.1} \right)^3 \log_{10} \left(\frac{M_0}{M_{in}} \right) \approx \begin{cases} 10 & M_0/M_{in} = 10^6 \\ 3.3 & M_0/M_{in} = 100 \\ 1.7 & M_0/M_{in} = 10 \end{cases}$$

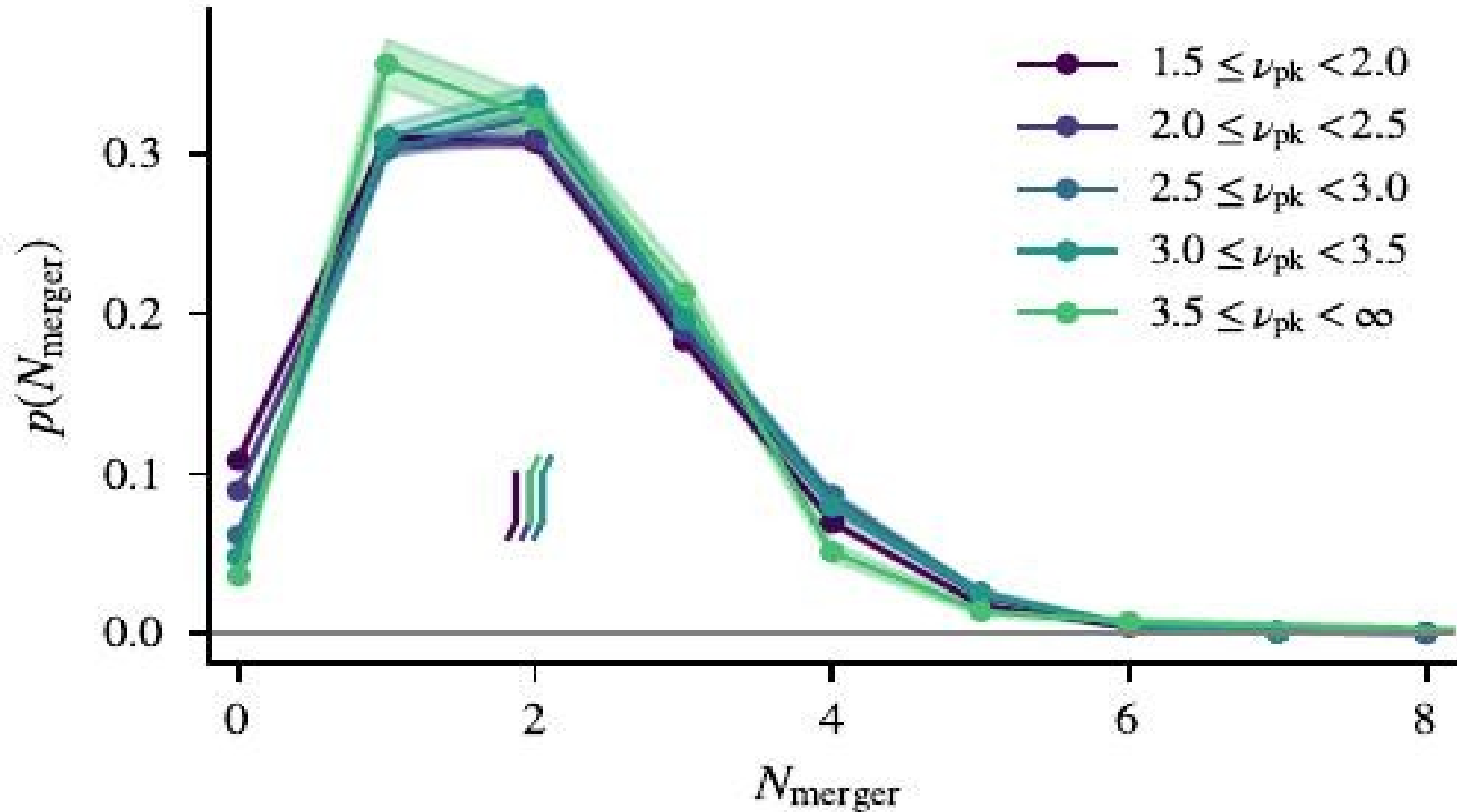
True mean radial number density of connected critical events with distance from central peak in a slab $d\ln R$ at fixed smoothing scale R
Numerical measure from skeleton reconstruction and finding connectors



Most connect critical events lie at near $r \approx 3R$ distance from peak,
 But it is the full integral of the curve that gives the mean merger count

Probability distribution of merger numbers during 10-fold mass increase

Numerical measure on initial $n=-2$ field from skeleton reconstruction



Average $N_{\text{merger}} \approx 2$ for $n=-2$ (LCDM-like) with weak dependence on peak height and very close to rough theoretical estimate !

Conclusions

- Since initial conditions for structure formation are given by a single scalar field of initial density, it is not surprising that some parameters of the Large Scale Structure can be deduced directly from it
- Some estimates are surprisingly accurate even in “zero order” approximation such as mean connectivity $k \approx 6$ or typical merger rate per decade of mass increase $N_{\text{merger}} \approx 2$. This was the focus of presentation.
- More detailed statistics, such as dependence of connectivity on cluster rarity, redshift evolution due to developing of non-Gaussianity or spin accumulation brought in by mergers, can also be understood ‘ab initio’ and is dependent on cosmology, through either change in power spectrum, and/or redshift timing of developing non-Gaussianity.
- Note that for the merger history I cared about peaks being physical collapsed halos, but not that filaments have appreciable density. But if you consider only physically dense filaments, the connectivity changes, and new concept of **halo multiplicity** μ (how many physical arms spread out from a cluster) arises as well. The answer is $\mu \approx 4$, on average
- There are still many avenues open to explore the features of filamentary network and critical events theory.